PREDICTING CUSTOMER SENTIMENT FROM TWITTER POSTS

APPLE AND GOOGLE PRODUCT SENTIMENT ANALYSIS

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Phase 4 Project

Overview

- Apple and Google products generate thousands of online discussions.
- Understanding customer sentiment manually is difficult.
- This project uses machine learning to classify tweets as:
  - Positive
  - Neutral
  - Negative
- The goal is to help organizations better understand customer opinions and identify areas requiring attention.

Business Problem

- Companies need to understand how customers feel about their products.
- Reviewing thousands of tweets manually is time-consuming.
- Negative sentiment may indicate product issues or customer dissatisfaction.
- The project aims to automate sentiment classification using machine learning.

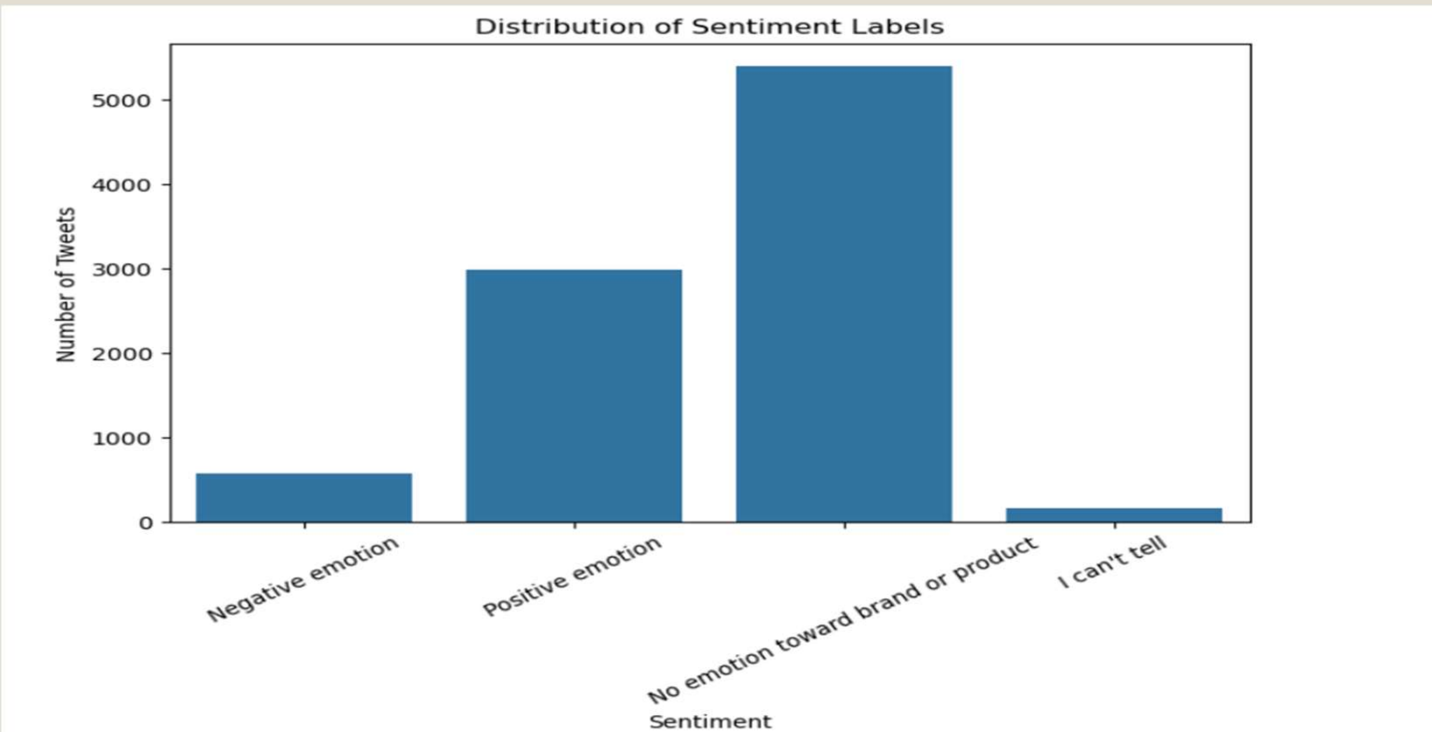
➢ **Stakeholders**

✓ Marketing Teams
✓ Product Development Teams
✓ Customer Experience Teams

Data Understanding

- Dataset contains 8,914 tweets.
- Tweets discuss Apple and Google products.
- Each tweet labeled as Positive, Neutral, or Negative.
- Dataset cleaned before analysis.
- Key Observation: Neutral tweets represent the largest class.

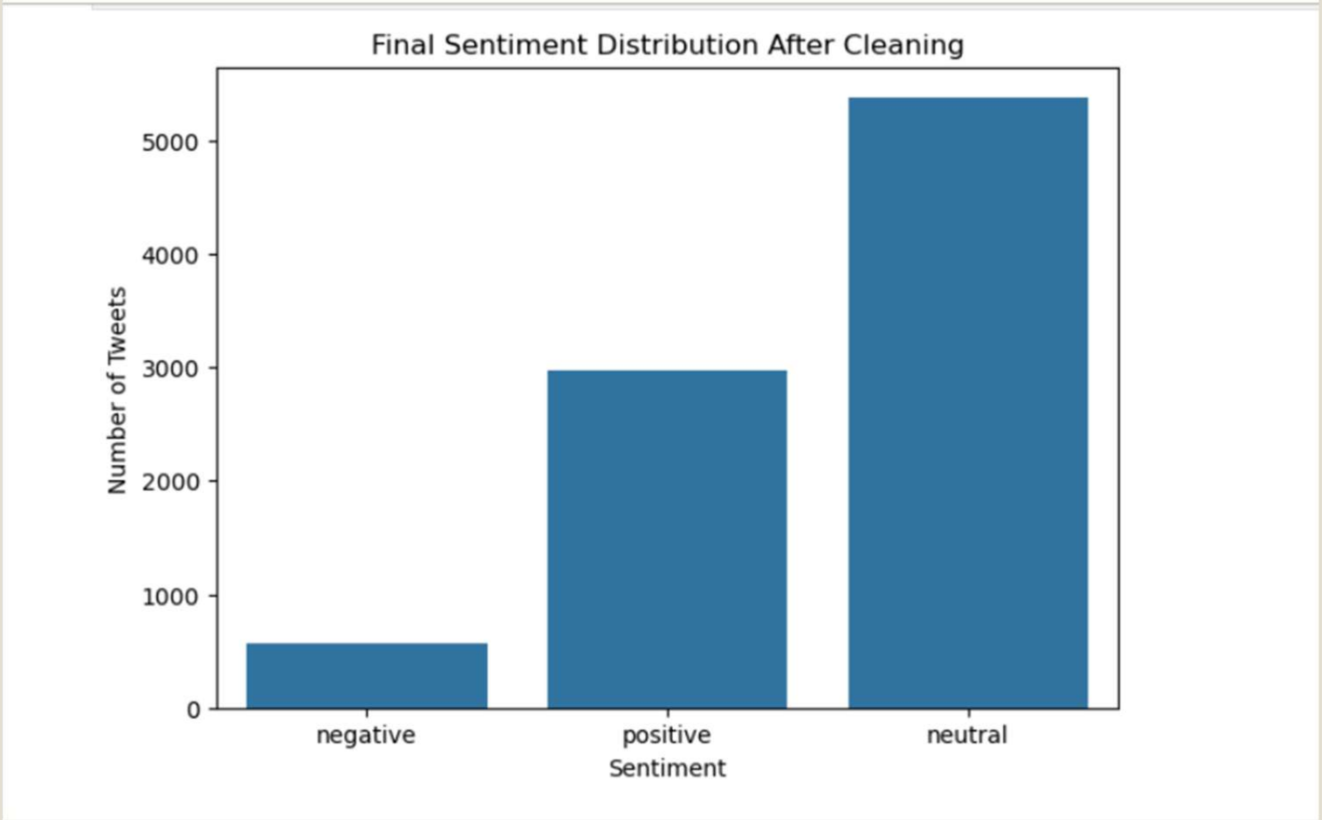
Data Understanding: Sentiment distribution chart



Data Preparation

- Removed missing values and duplicates.
- Cleaned tweet text.
- Removed URLs and punctuation.
- Converted text into numerical features using TF-IDF.
- Prepared data for machine learning models.

Final Sentiment distribution chart



Modelling Approach

- ◦ Why Classification?

❖Classification predicts tweet sentiment.

- Models Tested:

❖ Dummy Classifier

❖ Logistic Regression

❖ Decision Tree

❖ Tuned Decision Tree

❖ Random Forest

- Goal:

◦ ❖ Identify the model that best classifies tweet sentiment.

Logistic Regression Results

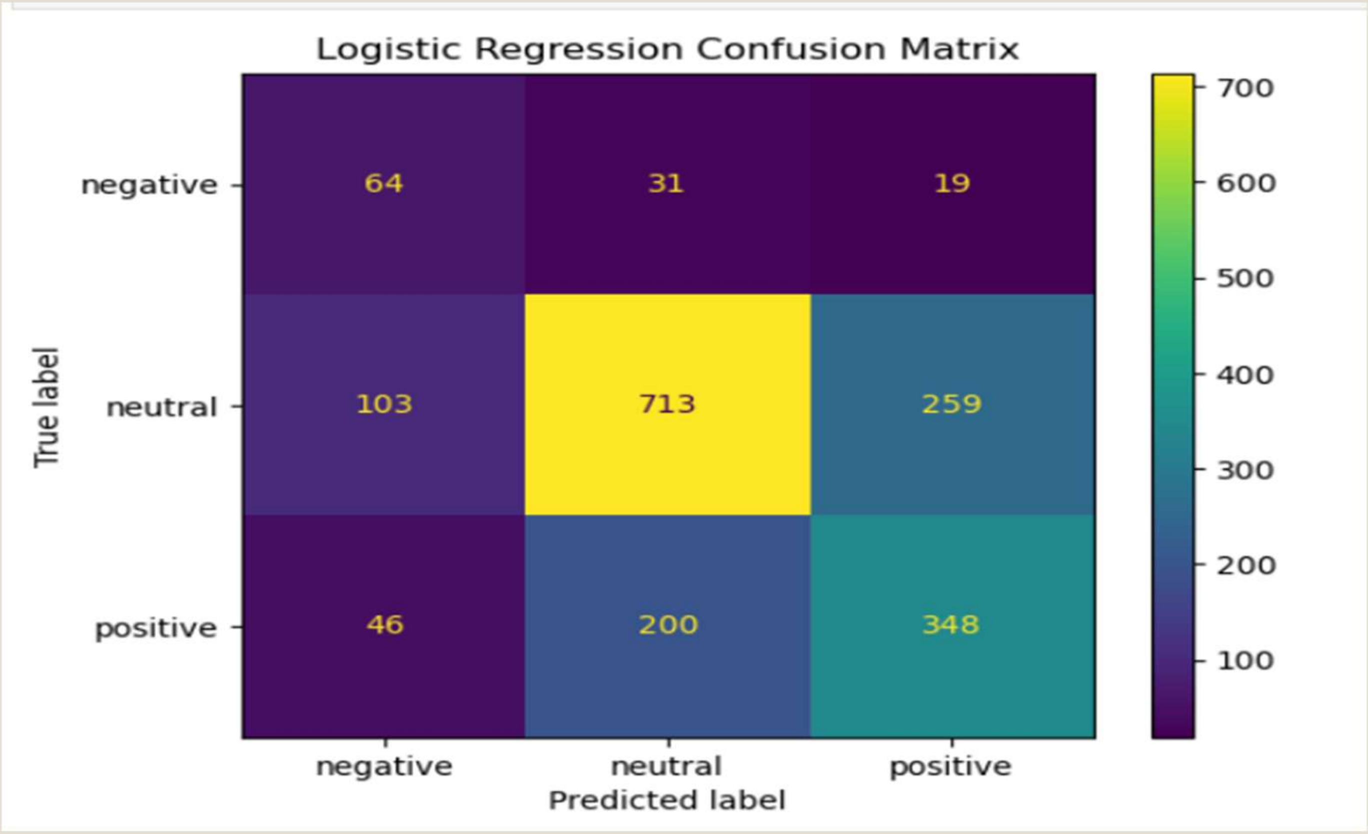
- Accuracy: 63.1%
- Macro F1-Score: 0.56
- Negative Recall: 56.1%

Why It Performed Well:

❖ Balanced performance across sentiment classes.

❖ Strong detection of negative tweets.

Logistic Regression Results



Model Comparison

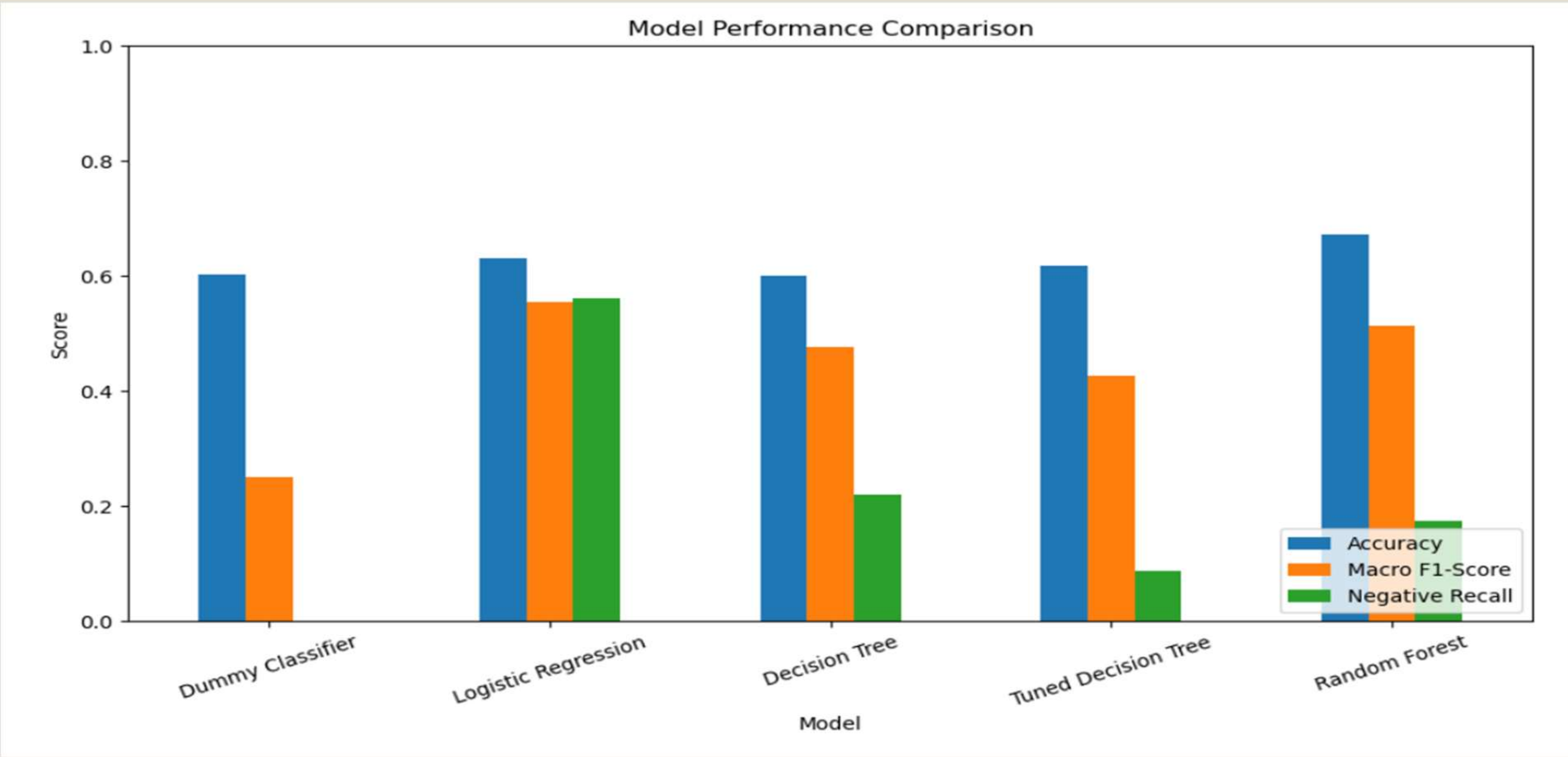
Model	Accuracy	Macro F1
Dummy	0.603	0.251
Logistic Regression	0.631	0.556
Decision Tree	0.600	0.478
Tuned Decision Tree	0.618	0.426
Random Forest	0.672	0.515

Conclusion:

• Logistic Regression achieved the best overall balance.

• Selected as final model.

Model Performance Comparison



Key Insights

Important Findings

❖ Positive tweets often contained words such as:

- great
- awesome
- love

❖ Negative tweets often contained words such as:

- fail
- headache
- hate

❖Business Meaning

✓Social media sentiment provides valuable customer feedback.

✓Negative sentiment can highlight customer concerns and product issues.

Business Recommendations

- Monitor social media sentiment regularly.
- Investigate recurring negative feedback.
- Use sentiment trends to support product improvements.
- Track customer reactions after product launches.

Next Steps

- Expand the dataset with newer tweets.
- Explore advanced NLP models such as BERT.
- Integrate sentiment monitoring into business dashboards.
- Continuously retrain the model with updated data.

Thank You!